Automated Irregular Shaped Object Stacking via Monte Carlo Tree Search and Optimization

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***Abstract*—This paper presents a novel approach for automated stacking of irregularly shaped objects. The proposed method combines Monte Carlo Tree Search (MCTS) with optimization techniques to achieve stable and efficient stacking configurations.**

I. Introduction

Automating the process of stacking objects has long been studied in the context of packing and warehousing systems. Our contributions are as follows:

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The rest of the paper is organized as follows. Sec. III Sec. V Sec. **??** Sec. VII Sec. VIII

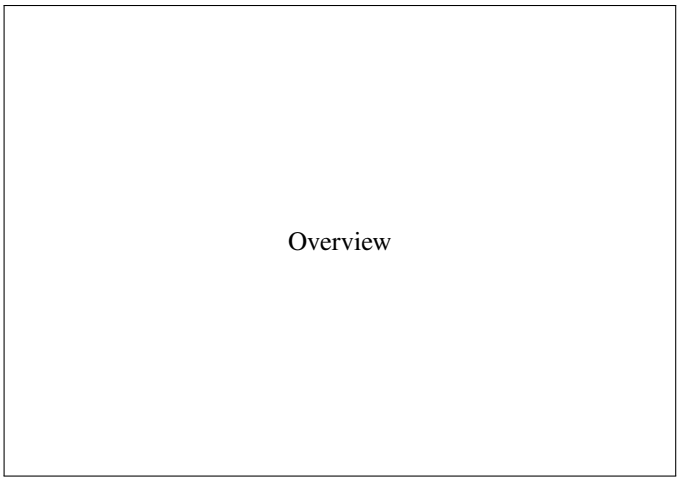


Fig. 1: Overview.

II. Related Works

III. Overview

A. Stacking as an Optimization Problem

Suppose that we are given a target shape $T \subset \mathbb{R}^3$ and n irregular shaped objects $O_1(q_1), O_2(q_2), \cdots, O_n(q_n) \subset \mathbb{R}^3$ where each $q_i \in \mathrm{SE}(3)$ represents the pose of each object O_i in the world frame. The goal is to find optimal placement poses $q_1, \cdots, q_n$ for a set of irregularly shaped objects such that the final configuration approximates the target shape T while satisfying the stability and non-collision constraints. This problem can be formulated as an optimization problem:

$$\min_{q_1,\cdots,q_n \in \mathrm{SE}(3)} \mathrm{Vol}\left(T \backslash \bigcup_{i=1}^{n} O_i(q_i)\right) \quad \text{s.t.} \quad c_{\text{phys}}(O_1(q_1), \cdots, O_n(q_n)) = 0 \tag{1}$$

where $\mathrm{Vol}(\cdot)$ is the volume of a 3D shape and $c_{\text{phys}}(\cdot)$ is a function that evaluates the physical feasibility of the configuration, including stability and non-collision constraints.

B. Markov Decision Process Formulation

Nevertheless, solving the optimization problem in Eq. (1) is extremely challenging due to the high-dimensional search space as the number of the objects n increases. In addition, the physical constraints involve complex contact dynamics which makes the problem even more difficult to solve.

To address the problem Eq. (1) in a tractable manner, we model the problem as a sequential decision-making process where we place one object at a time. Suppose that we have already placed k objects in the scene and denote the set of indices of the placed objects as $\mathcal{P}_k \subset \{1, \cdots, n\}$. Then, the k-th scene $\mathcal{X}_k := \{(i, q_i) \mid i \in \mathcal{P}_k\}$ is the collection of object indices and their corresponding placement poses that have been placed in the scene. Also, let the k-th inventory $\mathcal{I}_k := \{1, \cdots, n\} \backslash \mathcal{P}_k$ be the set of remaining object indices that have not been placed yet. Given $(\mathcal{X}_k, \mathcal{I}_k, T)$, we now need to select an index $i_{k+1} \in \mathcal{I}_k$ and find the corresponding placement pose q_{k+1} for the object such that the new scene $\mathcal{X}_{k+1} := \mathcal{X}_k \cup \{(i_{k+1}, q_{k+1})\}$ is physically feasible and approximates the target shape T.

The sequential decision-making process can be further modeled as a Markov Decision Process (MDP). We define the state space $\mathcal{S}$ as the set of states $s_k := (\mathcal{X}_k, \mathcal{I}_k, T)$, the action space $\mathcal{A}$ as the set of actions $a_k := (i_{k+1}, q_{k+1})$ representing the object index i_{k+1} and its placement pose q_{k+1}, the transition function $p : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ that maps a state-action pair to the next state, and the reward function $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ that evaluates the improvement in approximating the target shape T while satisfying the physical constraints. We will describe the details of the reward function design in the later section IV. Therefore, the MDP can be defined as a tuple:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, p, r, \gamma)$$

where $\gamma \in [0, 1]$ is the discount factor that determines the importance of future rewards.

C. Shape Representation

Throughout the paper, we use the union of differentiable support function (DSF) to represent the 3D shapes of the target shape T and the objects O_i, $i \in \mathcal{I}_k$, in the inventory. Given a compact convex body $C \subset \mathbb{R}^3$, the support function $h_C : \mathbb{R}^3 \backslash \{0\} \to \mathbb{R}$ is defined as

$$h_C(x) = \max_{s \in C} x \cdot s$$

where $\cdot$ denotes the standard inner product in $\mathbb{R}^3$. The work of Lee et al. [8] presented a differentiable version of the support function h_C that represents a smooth shape C.

The function $h_C(x)$ can be easily extended through rigid-body transformation as follows:

$$h_{\bar{C}(q)}(x) = h_C(R^\top x) + t \cdot x$$

for $q := (R, t) \in \mathrm{SE}(3)$ where $\bar{C}(q) = \{Rp + t \mid p \in C\}$ denotes a transformed C under rigid-body transformation. Given two DSF represented convex bodies $C_1(q_1)$ and $C_2(q_2)$ with SE(3) poses q_1, q_2, the signed distance $\mathrm{d}(C_1(q_1), C_2(q_2))$ between the two convex bodies can be easily computed using Riemannian optimization techniques associated with $h_{C_1(q_1)}$ and $h_{C_2(q_2)}$ as presented in An, et al. [2]. Moreover, due to the smoothness of DSFs, the signed distance $\mathrm{d}(C_1(q_1), C_2(q_2))$ and the other contact information (e.g., normal vectors and contact points) are differentiable with respect to q_1 and q_2, which allows us to use gradient-based optimization techniques.

D. Overall Structure of the Sequence Planner

The overall structure of the sequence planner is illustrated in Fig. 3. The sequence planner basically use Monte Carlo Tree Search (MCTS) to explore the space of possible stacking sequences, and during the search, it utilizes several components to sample the valid poses and evaluate the feasibility and quality of each candidate stacking sequence. The main components of the sequence planner include initial pose sampling module π_{init}, single object pose optimization module π_{opt}, and structural simulation module π_{simul}. We will further describe each component in detail in the later sections V and IV.

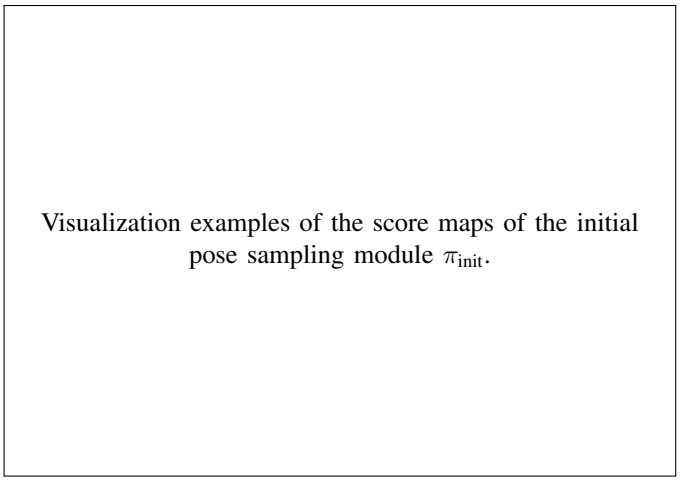


Fig. 2: Score maps of the initial pose sampling module π_{init}.

IV. Sequence Planning

In this section, we introduce the initial pose sampler π_{init} that generates initial poses for the pose optimization module π_{opt}, the structural simulation module π_{simul} that evaluates the stability of the scene, and the Monte Carlo Tree Search (MCTS) module π_{mcts} for sequence planning.

A. Initial Pose Sampling

The initial pose sampler $\pi_{\text{init}}(i_{k+1}, \hat{q}_{k+1} \mid s_k)$ generates candidate poses and corresponding object indices $i \in \mathcal{I}_k$ from the current state $s_k = (\mathcal{X}_k, \mathcal{I}_k, T)$ and the candidate pose $\hat{q}_{k+1}$ is refined by the pose optimization module π_{opt}. The sampler is formulated as the following product of two distributions:

$$\pi_{\text{init}}(i_{k+1}, \hat{q}_{k+1} \mid s_k) = \pi_{xy}(\mathbf{x} \mid i_{k+1}, R, s_k) \times \pi_{\text{unif}}(i_{k+1}, R \mid s_k)$$

where $\pi_{\text{unif}}(i_{k+1}, R \mid s_k)$ is a uniform distribution over $\mathcal{I}_k$ and a pre-defined finite rotation set, and $\pi_{xy}(\mathbf{x} \mid i_{k+1}, R, s_k)$ is a distribution over planar translations $\mathbf{x} = (x, y)$ given the object i_{k+1} and its orientation R. The height z of the candidate pose $\hat{q}_{k+1} = (R, t)$ is heuristically determined by the scene's upper-surface so that the object is placed above the scene since the pose optimization module π_{opt} will refine the candidate pose to find a stable placement pose q_{k+1} that is in contact with the scene. We will omit the dependency of π_{xy} on s_k in the following subsections for convenience.

For the planar translation distribution $\pi_{xy}(\mathbf{x} \mid i_{k+1}, R, s_k)$, we define a proposal score $\sigma(\mathbf{x} \mid i_{k+1}, R, s_k)$ that approximately estimates the quality of the candidate pose $\hat{q}_{k+1}$ based on the geometric information of the scene $\mathcal{X}_k$, the target shape T, and the object i_{k+1}. Then, we define

$$\pi_{xy}(\mathbf{x} \mid i_{k+1}, R, s_k) \propto \exp\left(\sigma(\mathbf{x} \mid i_{k+1}, R, s_k)/\tau\right)$$

so that the sampler favors candidate poses with higher proposal scores where $\tau > 0$ is a temperature parameter. The proposal score $\sigma(\mathbf{x} \mid i_{k+1}, R, s_k)$ is formulated as a weighted sum of several geometric features that are computed based on the scene $\mathcal{X}_k$, the target shape T, and the object i_{k+1}:

$$\sigma(\mathbf{x} \mid i_{k+1}, R) = w_{\text{s}}\sigma_{\text{s}}(\mathbf{x} \mid i_{k+1}, R) + w_{\text{f}}\sigma_{\text{f}}(\mathbf{x} \mid i_{k+1}, R) + w_{\text{b}}\sigma_{\text{b}}(\mathbf{x} \mid i_{k+1}, R) - w_{\text{h}}\sigma_{\text{h}}(\mathbf{x} \mid i_{k+1}, R),$$

where σ_{s} is the support area score map, σ_{f} is the filled floor area score map, σ_{b} is the boundary proximity score map, and σ_{h} is the height score map, respectively with weight parameters w_{h}, w_{s}, $w_{\text{f}}, w_{\text{b}} > 0$.

Intuitively, the score σ_{s} encourages the object to be placed on a larger support area, the score σ_{f} encourages the object to fill the empty floor area, the score σ_{b} encourages to fill the boundary of the target shape T first, and the last score σ_{h} encourages the object to be placed at a lower height to improve stability. The visualization examples of the score maps are shown in Fig. 2 and the details of the score map computation will be described in the appendix.

B. Structural Simulation

After the pose optimization module π_{opt} refines the candidate pose $\hat{q}_{k+1}$ to find a locally optimal placement pose q_{k+1}, we need to evaluate whether the new scene $\mathcal{X}_{k+1} := \mathcal{X}_k \cup \{(i_{k+1}, q_{k+1})\}$ is physically feasible and stable. The structural simulation module π_{simul} is responsible for simulating the physical behavior of the stacked objects under the influence of gravity and contact forces. For a given scene $\mathcal{X}_k$ and an action $a_{k+1} = (i_{k+1}, q_{k+1})$, the structural simulation module π_{simul} evaluates whether the new scene $\mathcal{X}_{k+1} := \mathcal{X}_k \cup \{(i_{k+1}, q_{k+1})\}$ is physically feasible and stable. Unlike the previous approaches of Johns et al. [6], [7] which also use the simulation to validate the stability of individual objects, we simulate the overall scene $\mathcal{X}_{k+1}$ instead of simulating only the newly placed object $O_{i_{k+1}}$ to ensure that the entire scene is stable with the GPU-accelerated subADMM algorithm [5] which is a scalable and efficient physics simulation algorithm for large-scale rigid body systems with contact.

C. Monte Carlo Tree Search for Sequence Planning

V. Pose Optimization

In this section, we describe the pose optimization module π_{opt} and the simulation module π_{simul} that refine a candidate placement pose to find a locally optimal configuration and evaluate whether the new scene is stable after placing a new object.

A. Single Object Pose Optimization Problem

For a given scene $\mathcal{X}_k$, a target shape T, and a candidate object $O_{i_{k+1}}$, each pose refinement can be formulated as solving a constrained optimization problem:

$$\begin{aligned} \min_{q,\lambda} \quad & [f_{\text{geo}}(q) + f_{\text{phys}}(q, \lambda)] \\ \text{s.t.} \quad & g_{\text{cone},j}(q, \lambda_j) \le 0, \quad j = 1, \cdots, n_c \end{aligned} \tag{2}$$

where $q \in \text{SE}(3)$ is the pose of the object and $\lambda = (\lambda_1, \cdots, \lambda_{n_c}) \in \mathbb{R}^{3n_c}$ is the vector of contact forces at the n_c contact points between the object and the scene.

The geometric objective $f_{\text{geo}}(q)$ is designed to evaluate how well the configuration approximates the target shape T while satisfying the physical constraints such as collision avoidance of $O_{i_{k+1}}$ in the scene $\mathcal{X}_k$. The detailed formulation of $f_{\text{geo}}(q)$ will be described in the appendix. The physical objective $f_{\text{phys}}(q, \lambda)$ is designed to evaluate whether the contact forces λ are physically feasible and contribute to a stable configuration. For indices $l, m \in \mathcal{P}_k$ such that $l < m$, let $I_{l,m} \subset \{1, \cdots, n_c\}$ be the set of contact point indices associated with the objects O_l, O_m, and let $I_l := \bigcup_{m \in \mathcal{P}_k} I_{l,m}$ be the set of contact point indices associated with the object O_l. Then, the physical objective $f_{\text{phys}}(q, \lambda)$ is defined as the following:

$$\begin{aligned} f_{\text{phys}}(q, \lambda) = w_{\text{eq}} \sum_{l \in \mathcal{P}_k} \| J_{c,I_l}^{\top}(q) \lambda_{I_l} + \lambda_{\text{ext},l} \|^2 \\ + w_{\text{cp}} \sum_{j \in I_{i_{k+1}}} \mathrm{d}_j(q)^2 \|\lambda_j\|^2 + \rho \sum_{j=1}^{n_c} \|\lambda_j\|^2 \end{aligned} \tag{3}$$

where $J_{c,I_l}(q) \in \mathbb{R}^{3|I_l| \times 6}$ is the Jacobian of the contact constraints associated with the object O_l at the contact points in I_l, $\lambda_{I_l} \in \mathbb{R}^{3|I_l|}$ is the vector of contact forces at the contact points in I_l, and $\lambda_{\text{ext},l} \in \mathbb{R}^6$ is the external wrench acting on the object O_l. The first term in $f_{\text{phys}}(q, \lambda)$ penalizes the violation of the static equilibrium constraints for each object in the scene. The second term penalizes the complementarity conditions on the contact forces between the candidate object $O_{i_{k+1}}$ and the objects in the scene $\mathcal{P}_k$ based on the signed distance $\mathrm{d}_j(q)$ between the two objects at the j-th contact point. The third term with $\rho > 0$ is a regularization term to enhance the convexity of the objective function. The weights $w_{\text{eq}}, w_{\text{cp}} > 0$ are parameters that balance the importance of each term in the force objective f_{phys}.

Finally, each $g_{\text{cone},j}(\cdot)$ is the smoothed friction cone constraints for the j-th contact point defined by

$$g_{\text{cone},j}(q, \lambda_j) = \sqrt{\|J_{c,j}^t \lambda_j\|^2 + \epsilon} - \mu_j J_{c,j}^n \lambda_j \tag{4}$$

where $\epsilon > 0$ is a smoothing parameter, μ_j is the friction coefficient, and $J_{c,j}^t$ and $J_{c,j}^n$ are the tangential and normal components of the Jacobian of the contact constraint at the j-th contact point.

We note that, due to the smoothness of DSFs, the terms in the objective function dependent to q are formulated which allows us to use gradient-based optimization techniques to optimize the placement pose of the object $O_{i_{k+1}}$.

B. Bi-level Optimization Formulation

Given a fixed pose q, the contact forces λ can be efficiently optimized by solving a quadratic programming problem under the second-order cone constraints. Therefore, rather than solving (2) directly, we can reformulate the problem as a bi-level optimization problem where the upper-level problem optimizes q and the lower-level problem optimizes λ for a given pose q:

$$\begin{aligned} \min_{q} \quad & \left[f_{\text{geo}}(q) + \min_{\lambda} f_{\text{phys}}(q, \lambda) \right] \\ \text{s.t.} \quad & g_{\text{cone},j}(q, \lambda_j) \leq 0, \quad j = 1, \cdots, n_c. \end{aligned} \tag{5}$$

Toy example of comparing the overall structural stability of the scene and the local structural stability of the candidate object

Fig. 3: Toy example of comparing the overall structural stability of the scene and the local structural stability of the candidate object.

Reflecting the definitions of $f_{\text{phys}}(q, \lambda)$ and $g_{\text{cone},j}(q, \lambda_j)$ in (3) and (4), we can easily see that the lower-level problem is a convex optimization problem with respect to λ for a fixed q which can be efficiently solved.

Denoting $\lambda^*(q)$ as the optimal λ of the lower-level problem for a given pose q, we define the force dependent objective function as $f_{\text{phys}}(q) := f_{\text{phys}}(q, \lambda^*(q))$. Then, using the *Envelope theorem* [3], we can compute the gradient of $f_{\text{phys}}(q)$ with respect to q as follows:

$$\frac{df_{\text{phys}}(q)}{dq} = \frac{\partial f_{\text{phys}}(q, \lambda^*)}{\partial q} + \sum_{j=1}^{n_c} \nu_j^* \frac{\partial g_{\text{cone},j}(q, \lambda_j^*)}{\partial q}$$

where ν_j^* is the optimal dual variable associated with the j-th friction cone constraint $g_{\text{cone},j}(q, \lambda_j^*) \leq 0$. Using the above gradient, we optimize the placement pose q on SE(3) manifold based on the Riemannian trust-region method [1] combined with Riemannian BFGS Hessian approximation [4] to find a locally optimal placement pose $q_{i_{k+1}}$ that minimizes the objective function $f_{\text{geo}}(q) + f_{\text{phys}}(q)$ while satisfying the friction cone constraints $g_{\text{cone},j}(q, \lambda_j^*(q)) \leq 0$. In addition, motivated by homotopy based optimization methods [], we designed the sequence of the weight parameters in $f_{\text{geo}}(q)$ and $f_{\text{phys}}(q)$ such as w_{eq}, w_{cp} and iteratively solve the bi-level optimization problem (5) to induce a better local minimum that satisfies the physical constraints and approximates the target shape T.

C. Force Optimization

As mentioned before, given a fixed pose q, the contact forces λ can be efficiently optimized by solving a quadratic programming problem under the second-order cone constraints. Moreover, exploiting the inherent structure of the problem, we solve the problem in scalable manner by applying an Alternative Direction Method of Multipliers (ADMM) []. Observe that each λ_j acts on two objects O_l and O_m such that $j \in I_{l,m}$ for some $l, m \in \mathcal{P}_k$ and $l < m$. Then, the

contact force λ_j can be split into two variables $\lambda_{j,1}$ and $\lambda_{j,2}$ such that $\lambda_{j,1} = \lambda_{j,2}$ where $\lambda_{j,1}$ is the contact force acting on O_l and $\lambda_{j,2}$ is the contact force acting on O_m. Extending this to all contact forces, we define the extended contact forces $\tilde{\lambda} = (\lambda_{1,1}, \lambda_{1,2}, \cdots, \lambda_{n_c,1}, \lambda_{n_c,2})$ and the index set $\tilde{I}_{l,m} := \{(j,s) \mid j \in I_{l,m}, s = 1 \text{ if } l < m, s = 2 \text{ if } l > m\}$ with $\tilde{I}_l := \bigcup_{m \in \mathcal{P}_k} \tilde{I}_{l,m}$. Then, the first term in $f_{\text{phys}}(q, \lambda)$ can be reformulated as

$$\sum_{l \in \mathcal{P}_k} \|J_{c,\tilde{I}_l}^\top(q)\tilde{\lambda}_{\tilde{I}_l} + \lambda_{\text{ext},l}\|^2$$

where $\tilde{\lambda}_{\tilde{I}_l} \in \mathbb{R}^{3|\tilde{I}_l|}$ is the vector of extended contact forces at the contact points in $\tilde{I}_l$. The important observation is that the reformulated term is separable with respect to each object O_l in the scene $\mathcal{P}_k$ since $\tilde{I}_l \cap \tilde{I}_m = \emptyset$ for any $l \neq m$. Then, the lower-level problem can be reformulated as the following optimization problem with the separable objective function and the consensus constraints:

$$\begin{aligned} \min_{\tilde{\lambda}} \quad & \tilde{f}_{\text{phys}}(q, \tilde{\lambda}) + \sum_{\substack{j=1,\cdots,n_c \\ s=1,2}} \delta_{K_j}(z_{j,s}) \\ \text{s.t.} \quad & \lambda_{j,1} = \lambda_{j,2} = z_{j,1} = z_{j,2} \\ & \text{for } j = 1, \cdots, n_c. \end{aligned}$$

where $\tilde{f}_{\text{phys}}(q, \tilde{\lambda})$ is the extended physical force function and $\delta_{K_j}(z_{j,s})$ is the indicator function of the second-order cone $K_j := \{\lambda \in \mathbb{R}^3 \mid g_{\text{cone},j}(q, \lambda) \leq 0\}$ for the j-th contact point and $z_{j,s}$ is an auxiliary variable for the j-th contact point and $s = 1, 2$. Applying ADMM to the above problem, we can decompose the problem into $|\mathcal{P}_k|$ independent subproblems for each object O_l in the scene $\mathcal{P}_k$, $2n_c$ times of projection onto the second-order cones K_j for $j = 1, \cdots, n_c$ which can be computed in closed-form and $2n_c$ times of averaging for the auxiliary variables $z_{j,s}$ for $j = 1, \cdots, n_c$ and $s = 1, 2$. The detailed ADMM algorithm for the lower-level problem is described in the appendix.

We want to ephasize that the force optimization approach considers the overall structural stability of the scene by evaluating the static equilibrium constraints for all objects in the scene, rather than just the candidate object $O_{i_{k+1}}$. Moreover, ADMM allows us to solve the problem in a scalable manner even for large-scale problems by decomposing the problem into independent subproblems for each object in the scene.

VI. Implementation and Evaluation

VII. Experiments

VIII. Conclusion

Acknowledgments

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